WW-P-471a
SEPTEMBER 27, 1960
SUPERSEDING
Fed. Spec. WW-P-471
Sept. 27, 1945
(See 6.5)

FEDERAL SPECIFICATION

PIPE-FITTINGS: BUSHINGS, LOCKNUTS, AND PLUGS; BRASS OR BRONZE, IRON OR STEEL, AND ALUMINUM; (SCREWED); 125—150 POUNDS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers bushings, locknuts, and plugs, made of brass or bronze, iron or steel, or aluminum for use with pipe or regular 125-pound cast iron, brass or bronze, and 150-pound malleable-iron screwed pipe fittings having American Standard Pipe Threads (NPT) or American Standard Dryseal Pipe Threads (NPTF). For higher pressures, solid plugs (not cored), and face bushings should be used.

1.2 Classification.

1.2.1 *Types, styles, and classes.* — Bushings, locknuts, and plugs shall be of the following types, styles, and classes, as specified (see 6.2):

Type I. — Brass or Bronze.

Type II. — Iron or steel.
- Style A.—Uncoated.
- Style B.—Black Oxide.
- Style C.—Zinc Coated.
- Style D.—Cadmium Plated.

Type III. — Aluminum.
- Style A. — Anodized.
- Style B. — Nonanodized.

Class 1. — American Standard Pipe Threads (NPT).

Class 2. — American Standard Dryseal Pipe Threads (NPTF).

1.2.2 *Kinds, sizes, and dimensions.* — Bushings, locknuts, and plugs shall be of the kinds, sizes, and dimensions indicated in tables I and II, and figure 1, as specified (see 6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-A-268 — Aluminum Alloy 24S; Bars, Rods, and Wire — Rolled or Drawn.

QQ-A-365 — Aluminum Alloy Bar, Rod, and Wire, (Free-Machining), 2011.

QQ-A-411 — Aluminum Alloy Bars, Rods, and Wire; (Rolled, Drawn, or Cold-Finished, 1100.

QQ-I-652 — Iron Castings, Gray.

QQ-I-666 — Iron Castings, Malleable.

QQ-P-416 — Plating, Cadmium (Electrodeposited).

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QQ–S–633 — Steel Bars, Carbon, Cold Finished and Hot Rolled, (General Purpose).
QQ–S–681 — Steel; Castings.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 — Marking for Domestic Shipment (Civilian agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–3 — Packaging and Packing for Delivery of Valves, Fittings and Flanges (Except for Oxygen Systems).
MIL–A–8625 — Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL–C–13924 — Coating, Oxide, Black, for Ferrous Metals.
MIL–Z–17871 — Zinc-Coating (Hot-Dip Galvanizing).

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 — Marking for Shipping and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28 — Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

American Standards Association Inc., Standards:

B16.14–1949—(R 1953) Ferrous Plugs, Bushings, and Locknuts with Pipe Threads.
B16.15–1958—Brass or Bronze Screwed Fittings, 125 lb.

(Copies may be obtained from the American Standards Association, 70 East Forty-Fifth St., New York 17, N. Y.)

Uniform Freight Classification Rules.

(Copies may be obtained from the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

National Motor Freight Classification Rules.

(Copies may be obtained from the American Trucking Associations Inc., 1424 16th Street, NW, Washington, D. C.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Brass or Bronze.* — Brass and bronze fittings (type I) shall be either cast metal or wrought metal having a composition as specified (see 6.2) and indicated in table III.

3.1.2 *Iron or steel.* — Iron and steel fittings (type II) shall be in accordance with class 20 to class 30 inclusive of Federal Specification QQ–I–652 for gray cast iron; grade I or grade II of Federal Specification QQ–I–666 for malleable iron; class X, class 1, or

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class 2 of Federal Specification QQ–S–681, and Federal Specification QQ–S–633 for steel.

3.1.3 *Aluminum.* — Aluminum fittings (type III) shall be in accordance with Federal Specifications QQ–A–268, QQ–A–365, and QQ–A–411.

3.2 Fabrication.—Bushings, locknuts, and plugs may be fabricated by casting, forging, or machining.

3.3 Dimensions.—Bushings, locknuts, and plugs shall conform to the dimensions specified in tables I and II and figure 1 for the sizes specified (see 1.2.2).

3.4 Finish.

3.4.1 *Uncoated (type II, style A).* — Unless otherwise specified (see 3.4.2, 3.4.3, and 3.4.4), the pipe fittings shall be provided in the uncoated state.

3.4.2 *Black oxide (type II, style B).* — When specified (see 6.2), the iron and steel pipe fittings shall be provided with a black oxide finish in accordance with Military Specification MIL–C–13924.

3.4.3 *Zinc-coated (type II, style C).* — When specified (see 6.2), the iron and steel fittings shall be zinc-coated by the hot-dip process in accordance with Military Specification MIL–Z–17871. The zinc coating shall be applied prior to all threading.

3.4.4 *Cadmium-plated (type II, style D).* —When specified (see 6.2), the iron and steel fittings shall be cadmium-plated in accordance with Federal Specification QQ–P–416.

3.4.5 *Anodized (type III, style A).* — When specified (see 6.2), the aluminum fittings shall be anodized in accordance with Military Specification MIL–A–8625.

3.4.6 *Nonanodized (type III, style B).* — Unless otherwise specified (see 3.4.5), aluminum fittings shall be nonanodized.

3.5 Threading.—Unless otherwise specified (see 6.2), all threads shall be right-handed and shall be smooth, clean, and true to form. Threads other than the internal thread in eccentric bushings, shall be concentric with the axis of the fitting. The entering end of all internal threads shall be chamfered approximately to the major diameter of the thread at the face of the fitting, at an angle of ap-

TABLE I. — *Kinds, Sizes and Dimensions*

Fittings	Dimensions		
	Type I Brass or Bronze	Type II Iron or Steel	Type III Aluminum
Bushings:			
Face	ASA B16.15	ASA B16.14	ASA B16.14
Outside head and inside head	ASA B16.15	ASA B16.14	ASA B16.14
Eccentric	------	Table II	------
Locknuts	ASA B16.14*	ASA B16.14	ASA B16.14
Plugs:			
Square head	ASA B16.15	ASA B16.14	ASA B16.14
Bar or slotted head	------	ASA B16.14	ASA B16.14
Countersunk	ASA B16.15	ASA B16.14	ASA B16.14

* Bronze locknuts are not made to standard dimensions, but they may be specified in the standard sizes indicated.

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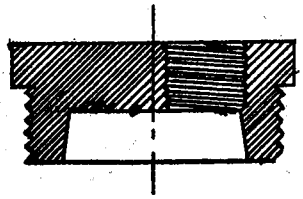


FIGURE 1.—Eccentric bushing.

TABLE II. — *Eccentric bushings*

Nominal Pipe Size		
Inches	*Inches*	*Inches*
1¼ x ¾	3 x 2	4 x 1½
1¼ x ½	3 x 1½	4 x 1¼
1½ x ¾	3 x 1¼	4 x 1
1½ x ½	3 x 1	5 x 4
2 x 1¼	3 x ¾	5 x 3½
2 x 1	3½ x 2½	5 x 3
2 x ¾	3½ x 2	5 x 2½
2 x ½	3½ x 1½	5 x 2
2½ x 1½	3½ x 1¼	6 x 4
2½ x 1¼	4 x 3	6 x 3
2½ x 1	4 x 2½	6 x 2½
2½ x ¾	4 x 2	6 x 2
2½ x ½		

Note 1.—Eccentric bushings are not made to standard dimensions, except as determined by standard thread dimensions for the sizes involved.

Note 2.—Heads of eccentric bushings shall be either hexagonal or octagonal, except that the larger sizes of eccentric bushings may have round heads with lugs.

proximately 45 degrees with the axis of the thread, and the entering end of all external threads shall be similarly chamfered approximately to the minor diameter of the thread. The chamfer shall be concentric with the thread, and shall be included in measurements of thread length. Threads shall be either American Standard Pipe Threads (NPT) or American Standard Dryseal Pipe Threads (NPTF) in accordance with National Bureau of Standards Handbook H28.

3.5.1 *Bushings.* — When specified (see 6.2 and 6.3), outside head bushings shall be externally threaded and gaged slightly oversized.

3.5.2 *Locknuts.* — All locknuts shall have internal American Standard Straight Pipe Threads (NPSL or NPSI) in accordance with National Bureau of Standards Handbook H28.

3.6 Identification marking.—The name or initial of the manufacturer, or other mark by which he can be readily identified, shall be legibly cast or die-stamped on each fitting, unless the size or shape does not permit such marking to be applied.

3.7 Workmanship.—Workmanship shall be in accordance with good commercial practice for this type of product.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities. — Unless otherwise specified herein (see 6.2), the supplier is responsible for the performance of all inspection requirements prior to submission for Government acceptance. Inspection records of the examinations and tests shall be kept complete, and shall be made available to the Government as specified (see 6.2). The supplier may utilize his own inspection facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.—A lot shall consist of all pipe-fittings of the same type, style, class, and size offered for delivery at one time.

4.3 Sampling.

4.3.1 *Sampling for examination.* — Sampling for examination shall be in accordance with Military Standard MIL–STD–105.

4.3.2 *Sampling for tests.* — Sampling for tests shall be in accordance with Military Standard MIL–STD–105.

4.4 Examination.—Each sample selected in accordance with 4.3.1 shall be visually ex-

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amined for the following defects to insure product conformance. Lot acceptance or rejection shall be based on an AQL of 2.5 percent defective for major defects and an AQL of 6.0 percent defective for minor defects.

4.4.1 *List of defects.*

Categories	*Defects*
Major:	
101.	Cold shuts.
102.	Blow holes.
103.	Porosity.
104.	Adherent sand or scale.
105.	Dimensions not within tolerances specified (see 4.4.2).
Minor:	
201.	Rough finish.
202.	Rough care.

4.4.2 Fittings in which limited small areas are slightly under the specified minimum metal thickness may be considered acceptable under this specification, provided such areas are at no point less than 90 percent of the specified thickness, and provided the fittings fully meet all other requirements.

4.5 **Tests.**—Each sample selected in accordance with 4.3.2 shall be tested as specified in 4.5.1, 4.5.2, 4.5.3, and 4.5.4. Lot acceptance or rejection shall be based on an AQL of 4.0 percent defective for tests specified in 4.5.1 and 4.5.2. The sample size for the tests specified in 4.5.3 and 4.5.4 shall be one unit from each lot. Failure to pass the test specified in either 4.5.3 or 4.5.4 shall be cause for rejection of the lot.

4.5.1 *Air-pressure test.* — Each fitting shall be subjected to an air pressure of not less than 60 pounds per square inch while the fitting is placed under water. The fitting shall not leak air.

4.5.2 *Hydrostatic-pressure test.* — Each fitting shall be subjected to a hydrostatic-pressure test of 250 pounds per square inch for one minute. The fitting shall not leak water.

4.5.3 *Composition test.* — Composition of the copper alloys shall be determined by spectrographic analysis or other suitable means of equivalent accuracy. The results of the analysis shall be within the limits specified in table III.

4.5.4 *Soundness test.* — Malleable-iron fittings shall be hammered or crushed to determine whether the heat treatment has been properly performed, and when zinc-coated, whether the coating is free from embrittlement. Failure to bend so that plastic deformation is readily apparent to the naked eye, or a showing of a typical malleable fracture, shall be considered evidence of unsatisfactory annealing, Flaking from the base metal during bending shall be evidence of embrittlement.

4.6 **Rejected lots.**—Disposition of rejected lots shall be in accordance with Military Standard MIL–STD–105.

4.7 **Defective units.**—Units found defective in the course of inspection shall be corrected to the satisfaction of the Government or shall be replaced with nondefective units.

4.8 **Inspection of preparation for delivery.** — The preservation, packaging, packing, and marking of the fittings shall be examined to determine compliance with the requirements of section 5 of this specification, and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with Military Standard MIL–STD–105, with inspection level L7, and an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing and equipment (for civil agencies only).

5.2 Packaging shall be level A or C as specified (see 6.2).

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TABLE III.—*Composition of brass or bronze*

Composition	1		2		2Wa		2Wb		3	
Element	Cast		Cast		Wrought		Wrought		Cast	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
	Percent	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*
Copper	86.0	90.0	84.0[1]	86.0	83.5	86.5	87.5	90.5	78.0	82.0
Tin	5.5	6.5	4.0	6.0	-----	-----	-----	-----	4.0	6.0
Lead	1.0	2.0	4.0	6.0	1.50	2.25	1.25	2.25	2.0	3.0
Zinc	3.0	5.0	4.0	6.0	Remainder		Remainder		10.0	16.0
Iron	-----	.25	-----	.30	-----	0.10	-----	0.10	-----	0.35
Nickel	-----	1.00	-----	1.00	-----	-----	-----	1.25	-----	0.75
Phosphorus	-----	.05	-----	.05	-----	-----	-----	-----	-----	0.05

[1] In determining compliance with the minimum copper requirement for composition 2 castings, copper may be computed as copper plus nickel.

5.2.1 *Level A.* — Fittings shall be packaged in accordance with the level A requirements of Military Specification MIL–P–3.

5.2.2 *Level C.* — Fittings shall be packaged in accordance with supplier's standard practice.

5.3 **Packing.**—Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 *Level A.* — Fittings shall be packed in accordance with the level A requirements of Military Specification MIL–P–3.

5.3.2 *Level B.* — Fittings shall be packed in accordance with the level B requirements of Military Specification MIL–P–3.

5.3.3 *Level C.* — The pipe fittings shall be packed to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules and regulations as applicable to the mode of transportation.

5.4 **Marking, civil agency.**—In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.5 **Marking, military.**—In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.**—Bushings, locknuts, and plugs are intended for use with threaded piping at working pressures between 125 and 150 pounds.

6.2 **Ordering data.** — Procurement documents should specify the following:

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(*a*) Title, number, and date of this specification.

(*b*) Type, style, kind, class, size and dimensions of fittings required (see 1.2).

(*c*) Composition of copper alloy required (see 3.1.1 and table III).

(*d*) Finish required (see 3.4).

(*e*) When left-hand threads are required (see 3.5).

(*f*) Whether outside head bushing is required and the dimensions of the oversize thread (see 3.5.1 and 6.3).

(*g*) Deviations, if any, from the inspection requirements, and how and where the inspection records are to be made available (see 4.1).

(*h*) Level of packaging and level of packing required (see 5.2 and 5.3).

6.3 Outside head bushing. — The outside head bushings are required in installations where the shoulder of the head would interfere in the makeup of the joint if a regular bushing were used.

6.4 Under the revised Federal Specification (WW–P–471a), different materials are designated by type, whereas differences in finish were designated by type (WW–P–471) prior to this revision. The different finishes are now designated by style.

6.5 Supersesion data.—This specification includes the requirements of Military Specifications MIL–B–20235, dated 5 November 1951; MIL–F–1184 dated 12 September 1949; and MIL–F–1181 dated 12 September 1949, in part.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy—Bureau of Yards and Docks
Air Force—Mobile Air Materiel Area

OTHER INTEREST:
Navy—Sh.

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